

Program-8

Title: Create Kubernetes Pods Using YAML Descriptors

1. Minikube setup and status check

Minikube is used here to run a single-node Kubernetes cluster locally.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg8$ minikube start
🐳 minikube v1.37.0 on Fedora 43
🌟 Using the qemu2 driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🔄 Updating the running qemu2 "minikube" VM ...

🔗 Configuring bridge CNI (Container Networking Interface) ...
🔍 Verifying Kubernetes components...
  ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: default-storageclass, storage-provisioner, dashboard
🏡 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
• rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg8$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

○ rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg8$
```

2. Application and Configuration Files

```
! nodejs-pod.yaml U X
! nodejs-pod.yaml
1  apiVersion: v1
2  kind: Pod
3
4  metadata:
5    name: nodejs-pod
6
7  spec:
8    containers:
9      - name: nodejs
10        image: node:18
11        command: ["node", "/usr/src/app/server.js"]
12        volumeMounts:
13          - name: app-code
14            mountPath: /usr/src/app
15        ports:
16          - containerPort: 3000
17
18    volumes:
19      - name: app-code
20        configMap:
21          name: nodejs-app-config
22
```

! nodejs-configmap.yaml U X

! nodejs-configmap.yaml

```
1  apiVersion: v1
2  kind: ConfigMap
3
4  metadata:
5    name: nodejs-app-config
6
7  data:
8    server.js: |
9      const http = require("http");
10
11      const server = http.createServer((req, res) => {
12        res.end("Hello from Node.js running inside Kubernetes!");
13      });
14
15      server.listen(3000, () => {
16        console.log("Server running on port 3000");
17      });
18
```

JS server.js X

JS server.js > ...

```
1  const http = require("http");
2  const port = 3000;
3
4  const server = http.createServer((req, res) => {
5    res.end("Hello from Node.js running inside Kubernetes!");
6  });
7
8  server.listen(port, () => {
9    console.log(`Server running at port ${port}`);
10  });
11
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg8$ kubectl apply -f ./nodejs-configmap.yaml
configmap/nodejs-app-config created
● rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg8$ kubectl apply -f ./nodejs-pod.yaml
pod/nodejs-pod created
● rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg8$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
nginx-deployment-6f9664446b-2599x   1/1     Running             3 (13m ago) 4h57m
nginx-deployment-6f9664446b-lbr2k   1/1     Running             3 (13m ago) 4h57m
nodejs-pod                           0/1     ContainerCreating   0           25s
● rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg8$ kubectl get configmaps
NAME      DATA  AGE
kube-root-ca.crt  1      23h
nodejs-app-config  1      79s
● rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg8$ kubectl logs nodejs-pod
Error from server (BadRequest): container "nodejs" in pod "nodejs-pod" is waiting to start: ContainerCreating
● rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg8$ kubectl port-forward pod/nodejs-pod 3000
:Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
```

Access the nodejs-app:

